

1. Give a *direct proof* of the following statements.

(a) Recall that, for natural numbers n and k with $k \leq n$, the binomial coefficient is defined as $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. Show that, if $n \geq 2$ is an integer, that $n^2 = 2\binom{n}{2} + \binom{n}{1}$.

(b) Let a and b be integers. If $a|b$, then $a|(3b^3 - 4b^2 + 18b + 2a)$.

(c) If x is an integer and $x^2|x$, then $x \in \{-1, 0, 1\}$.

2. Give a *contrapositive proof* of the following statements.

(a) Suppose x and y are integers. If $3 \nmid xy$, then $3 \nmid x$ and $3 \nmid y$.

(b) (Hammack Problem 5.6) Suppose x is a real number. If $x^3 - x > 0$, then $x > -1$.

(c) (Hammack Problem 5.7) Suppose a and b are integers. If both ab and $a + b$ are even, then both a and b are even.

Prove the following using *proof by contradiction*.

3. $\sqrt{2}$ is irrational.

4. There are infinitely many prime numbers.

5. (Basically Hammack Problem 6.16)¹ Suppose x and y are non-negative real numbers. Then

$$\frac{x+y}{2} \geq \sqrt{xy}$$

or, equivalently, $x+y \geq 2\sqrt{xy}$.

¹This is also called the AM-GM Inequality, as it says that the Arithmetic Mean $\frac{x+y}{2}$ is always greater than or equal to the Geometric Mean $\sqrt{xy}$.

Proof Methods – Solutions

1. Give a *direct proof* of the following statements.

- (a) Recall that, for natural numbers n and k with $k \leq n$, the binomial coefficient is defined as $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. Show that, if $n \geq 2$ is an integer, that $n^2 = 2\binom{n}{2} + \binom{n}{1}$.

Proof. This will be a direct proof. So, take $2\binom{n}{2} + \binom{n}{1}$ and substituting in for the binomial expressions, we get:

$$\begin{aligned} 2\binom{n}{2} + \binom{n}{1} &= 2\frac{n!}{2!(n-2)!} + \frac{n!}{1!(n-1)!} = 2\frac{n(n-1)}{2} + \frac{n}{1} = \\ &= n(n-1) + n = n^2 - n + n = n^2 \end{aligned}$$

□

- (b) Let a and b be integers. If $a|b$, then $a|(3b^3 - 4b^2 + 18b + 2a)$.

Proof. If $a|b$, then $b = ak$ for some integer k . So then we can substitute in for b in the expression and get:

$$3b^3 - 4b^2 + 18b + 2a = 3(ak)^3 - 4(ak)^2 + 18(ak) + 2a = a(3a^2k^3 - 4ak^2 + 18k + 2)$$

Note that $3a^2k^3 - 4ak^2 + 18k + 2$ is an integer, so $a|b$.

□

- (c) If x is an integer and $x^2|x$, then $x \in \{-1, 0, 1\}$.

Proof. If $x^2|x$, we have that $x = x^2k$ for some integer k . We have two cases. If x is not zero, then we can cancel an x and get that $xk = 1$, so $x|1$, so $x = \pm 1$. The other case is if $x = 0$. So $x \in \{-1, 0, 1\}$.

□

2. Give a *contrapositive proof* of the following statements.

- (a) Suppose x and y are integers. If $3 \nmid xy$, then $3 \nmid x$ and $3 \nmid y$.

Proof. The contrapositive statement is: if $3|x$ or $3|y$, then $3|xy$. So assume without loss of generality $3|x$, so $x = 3k$ for some integer k . So $xy = (3k)y = 3(ky)$. As ky is an integer, $3|xy$.

□

- (b) (Hammack Problem 5.6) Suppose x is a real number. If $x^3 - x > 0$, then $x > -1$.

Proof. The contrapositive statement is: if $x \leq -1$, then $x^3 - x \leq 0$. (Note here that the opposite of greater than is less than or equal to) If $x \leq -1$, then $x^3 \leq x$, so $x^3 - x \leq 0$.

□

- (c) (Hammack Problem 5.7) Suppose a and b are integers. If both ab and $a + b$ are even, then both a and b are even.

Proof. The contrapositive statement is: if a or b is odd, then $a + b$ or ab is odd. There are two cases here. The first case is if only one of them are odd. So assume without loss of generality that a is odd and b is even, so $a = 2k + 1$ and $b = 2j$ for integers j, k . Then, $a + b = 2k + 1 + 2j = 2(k + j) + 1$, which is odd.

The second case is if both are odd. So $a = 2k + 1$ and $b = 2j + 1$ for integers j, k . Then, $ab = (2k + 1)(2j + 1) = 4jk + 2j + 2k + 1 = 2(2jk + j + k) + 1$ is odd.

□

Prove the following using *proof by contradiction*.

3. $\sqrt{2}$ is irrational.

Proof. Assume towards a contradiction that $\sqrt{2}$ is rational. Then $\sqrt{2} = \frac{p}{q}$. We can assume that p and q have no common factors by reducing. So, squaring both sides gives us $2 = \frac{p^2}{q^2}$, so $2q^2 = p^2$, which means p^2 is even. However, this also means p is even. So, $p = 2k$ for some integer k , so $p^2 = 4k^2 = 2q^2$. This gives us that $2k^2 = q^2$, so q^2 is even, so q is even. However, this is a contradiction, as p and q are both even but have no common factors. $\square$

4. There are infinitely many prime numbers.

Proof. Suppose towards a contradiction that there are finitely many prime numbers $p_1, \dots, p_n$, with p_n the largest prime number. Consider the number $N = p_1 \dots p_n + 1$. This number is larger than our largest prime, so it can't be prime, so it must be divisible by some p_i , a prime. However, if p_i divides N and p_i divides $p_1 \dots p_n$, then p_i divides $N - p_1 \dots p_n = 1$, which is a contradiction. $\square$

5. (Basically Hammack Problem 6.16)² Suppose x and y are non-negative real numbers. Then

$$\frac{x+y}{2} \geq \sqrt{xy}$$

or, equivalently, $x + y \geq 2\sqrt{xy}$.

Proof. Suppose towards a contradiction that $x + y < 2\sqrt{xy}$. As x and y are non-negative, we can square both sides and get $(x + y)^2 < 4xy$ so $x^2 + 2xy + y^2 < 4xy$, so $x^2 - 2xy + y^2 < 0$, so $(x - y)^2 < 0$. However, this is a contradiction as no real number squared can be less than 0. $\square$

²This is also called the AM-GM Inequality, as it says that the Arithmetic Mean $\frac{x+y}{2}$ is always greater than or equal to the Geometric Mean $\sqrt{xy}$.